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Service pricing in resilient multimodal transportation networks

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ABSTRACT

We study a fourth-party logistics platform that aims to build a resilient multimodal transportation network. The platform does not operate its own physical assets but constructs the network by renting transportation resources from third-party logistics carriers and making pricing and capacity planning decisions. To address this problem, we develop a scenario-based stochastic programming model embedded with robust optimization, which seeks to maximize the platform's total profit under both demand and supply uncertainties. The model captures two key sources of uncertainty: (i) demand is modeled as a price-dependent function, where uncertainty is introduced via a demand scaled factor that is bounded by an uncertainty set; (ii) the remaining capacity on each transportation arc varies across a set of disruption scenarios, reflecting supply-side risks. To solve this model, we propose a customized column-and-constraint generation algorithm that addresses the violation of the relatively complete recourse assumption by integrating feasibility and optimality subproblems. To further enhance scalability for large instances, we develop a CPU-GPU hybrid heuristic that accelerates second-stage evaluations through parallelization. We conduct a series of computational experiments based on real data from southeastern China to evaluate model performance and derive managerial insights. Results show that the platform should pay attention to the impact of OD-level demand heterogeneity on profit. Ignoring capacity disruptions may lead to overly optimistic and fragile solutions, highlighting the importance of explicitly modeling disruption scenarios. Moreover, results provides guidelines for identifying key nodes that are critical to mitigating the impact of disruptions.

1. Introduction

With the expansion of global trade and rising customer expectations for timely and reliable delivery, logistics platforms face challenges in balancing cost efficiency with service quality. In long-distance, high-volume, and cross-regional transportation scenarios, customer demands are becoming more diverse, with a growing emphasis on service customization. Due to limitations in coverage, routing flexibility, and coordination capabilities, these needs are difficult to fulfill through single-mode transportation. As a result, multimodal transportation networks, which integrate truck, rail, and ship modes, have emerged as an effective solution for meeting complex transportation demands (Wu et al. 2023). Such networks offer advantages in cost-effectiveness, spatial coverage, and resource complementarity, and are becoming an essential mode for supporting integrated logistics operations. In this context, fourth-party logistics (4PL) platforms play a critical role as independent service coordinators, distinct from both customers and carriers. Rather than owning transportation assets, 4PL platforms rent transportation capacity from third-party logistics providers and preplan multiple

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feasible transportation plans between origin–destination (OD) pairs to enable flexible future freight matching. It is worth noting that under single-mode transportation, the 4PL platform has limited economic value because all feasible routes share similar cost structures, leaving little opportunity to design differentiated transportation plans or generate margin beyond what individual carriers already provide. In contrast, multimodal operations enable the design of substitutable and multi-layered transportation plans by combining truck, rail, and ship arcs. This heterogeneity creates meaningful cost differences across OD-specific plans. During operations, the platform must determine the service price for each OD pair, allocate demand across different transportation plans, and reserve capacity in advance. These pricing, capacity planning, and flow allocation decisions are interdependent and form the core of operational optimization for 4PL platforms.

In the real world, the transportation environment faced by 4PL platforms is full of uncertainties, mainly from two sources. On the supply side, extreme weather events such as highway speed restrictions and port closures caused by typhoons, may reduce or interrupt the available transport capacity, which then renders certain patterns or plans temporarily unavailable in specific areas. On the demand side, transportation demand is highly uncertain, which can lead to supply–demand imbalances. If only a single transportation mode is used, disruptions such as road closures, port shutdowns, or rail suspensions may directly sever the connection between origin and destination or require excessively long rerouting, which severely compromises service continuity. In contrast, multimodal networks can maintain connectivity by switching freight to alternative modes that remain functional under disruption conditions. This flexibility enhances the robustness of transportation services, but it also increases the complexity of transshipment arrangements and the coordination of heterogeneous resources, making operational planning more difficult for the platform. Therefore, although multimodal networks introduce greater operational complexity, their superior connectivity under uncertainty makes them indispensable for platform-based logistics systems. If the platform only aims to profit optimization, it often ignores the adaptability of the transportation network to sudden disturbances and then faces the risk of local service paralysis or customer loss. Building a resilient multimodal transportation network thus becomes a critical strategy for ensuring stable platform operations. Network resilience is not only reflected in the ability to maintain a basic service level after a disruption, but also in the platform's ability to flexibly adjust through pre-emptive transportation network design. This enables a structured trade-off between service reliability and economic efficiency. To support this goal, disruption scenarios must be incorporated into the modeling framework, and quantifiable resilience metrics should be introduced to systematically evaluate the platform's capacity to withstand shocks and to guide robust pricing and capacity planning decisions.

To address multi-source uncertainties and resilience requirements in multimodal transportation networks, this paper develops a systematic decision-making framework for 4PL platforms to jointly optimize transportation capacity planning, pricing, and resource allocation strategies. Specifically, multiple transportation plans are designed for each OD pair, and the platform is allowed to flexibly select and reserve transportation capacity at the arc level, enabling it to better respond to variations in resource availability under different disruption scenarios. Meanwhile, differentiated pricing is applied across OD pairs to align transportation services with customer demand heterogeneity, thereby improving profitability. We formulate a scenario-based stochastic programming model embedded with robust optimization to incorporate disruption scenarios and quantifiable resilience metrics, which systematically capture the impact of supply and demand uncertainties on network performance. We further develop a customized column-and-constraint generation (CCG) algorithm to efficiently solve the proposed model. Since the model does not meet the relatively complete recourse assumption, the algorithm introduces a tailored feasibility subproblem to detect disruption scenarios under which the pricing and capacity planning decisions fail to satisfy resilience constraints. Only when all scenarios are feasible is the optimality subproblem invoked to identify the worst-case capacity allocation that minimizes the platform's profit. In addition to the CCG algorithm, we design a CPU-GPU hybrid heuristic to enhance scalability when instances become large. The method decomposes the solution process into a CPU-based first-stage update and parallel second-stage evaluations on the GPU, enabling rapid assessment of large scenario sets through parallel computation. This hybrid design substantially reduces computational time and makes the framework applicable to realistic large-scale multimodal networks. Finally, computational experiments based on real-world data from southeastern China demonstrate the model's advantages in enhancing platform profitability, service reliability, and network resilience, highlighting its practical value in complex transportation system management.

The remainder of this paper is structured as follows. Section 2 reviews the related literature. Section 3 briefly describes the problem background. Section 4 presents the mathematical model, followed by the solution algorithm in Section 5. Section 6 reports the computational results and managerial insights. Finally, Section 7 concludes the paper with a summary of findings and future research directions.

2. Literature review

This paper develops an optimization framework for service pricing and capacity/resource allocation in transportation networks. In the aspect of modeling, a hybrid model that combines the scenario-based stochastic programming and the robust optimization is proposed in this paper. Accordingly, the related works are reviewed from the following two research streams. One is about the specific features in the problem modeling; the other is about the recent advances in modeling with the hybrid characteristic.

2.1. Pricing and allocation problems in transportation networks

Several studies on pricing and allocation problems in transportation networks have primarily focused on optimizing either pricing or allocation. For example, Alizadeh et al. (2013) extend a classical bilevel pricing model to a two-stage stochastic framework, capturing tariff decisions under inter-temporal consistency. Azadian and Murat (2018) address the group-to-group pricing problem under price-sensitive demand using a mixed-integer nonlinear model and decomposition-based exact solution algorithms. More recent

research has increasingly explored joint optimization of pricing and allocation problem. Wu et al. (2011) propose a Pareto-improving pricing and allocation model for mitigating congestion in multimodal networks. Wang and Meng (2021) investigate pricing decisions for each OD pair in liner container shipping under uncertain demand and residual capacity. A two-stage stochastic nonlinear non-convex model is proposed and globally solved using a spatial branch-and-bound-and-Benders algorithm. Babaee et al. (2022) propose an integrated pricing and demand allocation model for perishable supply chains, utilizing a Benders decomposition to improve profitability and reduce waste. Erdogan et al. (2022) develop a bilevel hub location and pricing model involving non-cooperative interactions, demonstrating how arc pricing affects network flow allocation. Xu et al. (2022) study an OD-specific pricing and seat allocation model for passenger rail systems under price-sensitive demand, solved via linearization-relaxation. Beliaev et al. (2024) constructed a congestion game model with heterogeneous users in multimodal pickup and delivery networks, deriving closed-form delivery prices for arbitrary allocation objectives. Han et al. (2025) incorporate customer segmentation into a multimodal platform, jointly optimizing transport planning, capacity allocation, and pricing to improve profitability under demand fluctuations. While the literature has advanced from pricing and allocation to joint optimization strategies, few studies address the integrated handling of pricing, capacity planning, allocation, supply and demand uncertainty, and network resilience under disruption environments. To ensure methodological consistency, the studies summarized in Table 1 are restricted to contributions within the operations research (OR) literature that jointly consider pricing and allocation in transportation networks. These papers represent the subset of OR-based models that are most comparable to our problem setting, particularly in terms of decision structure and analytical formulation. As summarized in Table 1, most previous studies on pricing and allocation assume deterministic or purely stochastic demand, where price and demand are modeled as independent or exogenous variables. In contrast, this study defines demand as a price-dependent and uncertain function, which allows endogenous interaction between pricing and allocation decisions and provides a more realistic representation of customer behavior in platform-based logistics systems. Moreover, unlike existing models that mainly focus on single-mode or static capacity settings, the proposed framework jointly considers multimodal transportation, capacity planning, and resilience constraints, capturing the platform's operational flexibility under potential disruptions. By embedding quantifiable resilience requirements into the pricing and allocation framework, the model ensures reliable service performance under capacity interruptions.

2.2. Modeling based on stochastic programming and robust optimization

This paper proposes a mixed-integer linear programming (MILP) model that combines scenario-based two-stage stochastic programming and robust optimization; this modeling idea is similar to the "two-stage stochastic robust model", which has been formulated in some related works during recent years. For example, Liu et al. (2019) develop a two-stage stochastic robust model for warehouse operations under uncertain product demand, with first-stage decisions on container procurement and product packing, and second-stage decisions on container stacking based on realized demand. Similarly, Wang et al. (2024) introduce a two-stage stochastic robust framework under binary uncertainty, where first-stage resource allocation is followed by adaptive adjustments based on disruption realizations. In both studies, scenario-based uncertainty is reflected in the construction of uncertainty sets, while other model parameters remain fixed across scenarios. However, subsequent studies go beyond merely varying the uncertainty sets and allow certain model parameters to vary across scenarios. Sun et al. (2022) propose a bi-objective scenario-based robust optimization model for humanitarian logistics, jointly optimizing medical facility locations, casualty transportation, and relief allocation under

Table 1
Comparison with related works on operation optimization in transportation networks.

Study	Demand modeling	Model	Multi-modal	Capacity planning	Resilience & service-level	Solution method	Uncertainty sources
Alizadeh et al. (2013)	Stochastic demand model	M				Commercial solver	Demand
Azadian and Murat (2018)	Deterministic price function	M				Benders and branch and bound algorithm	
Wu et al. (2011)	Deterministic model	G	✓		Constraints		
Wang and Meng (2021)	Price function with uncertainty	M				Branch-and-bound-and-benders algorithm	Demand and supply
Babaee et al. (2022)	Deterministic model	M			Constraints	Benders algorithm	
Erdogan et al. (2022)	Deterministic model	M				Benders algorithm	
Xu et al. (2022)	Deterministic price function	M				Customized global-optimization	
Beliaev et al. (2024)	Deterministic model	G	✓		Objective	Commercial solver	
Han et al. (2025)	Price function with uncertainty	M	✓	✓		Hybrid particle swarm algorithm	Demand
This paper	Price function with uncertainty	M	✓	✓	Constraints	Column-and-constraint generation	Demand and supply

Notes: The "Model" column classifies the modeling approach adopted in each study. "M" denotes mathematical programming models and "G" refers to game-theoretic models; For the "Resilience & service-level" column, "Objective" indicates that the study incorporates resilience or service levels as optimization objectives, while "Constraints" indicates their treatment as hard constraints.

facility disruptions and uncertain demand. Yang et al. (2024) formulate a hybrid two-stage stochastic robust model for emergency evacuation and blood product scheduling, considering both scenario-based disruptions and robust uncertainty in casualty demand, solved via a proximal bundle algorithm. Luo et al. (2025) address supply chain network design under disruption risk and endogenous demand uncertainty, using an adjusted CCG algorithm to solve the two-stage stochastic-robust model. Across these works, scenario-based uncertainty typically affects disrupted facilities, and crucially, all assume relatively complete recourse, ensuring second-stage feasibility under all scenarios. In contrast, Matthews et al. (2019) examine a two-stage robust optimization problem for resilient flow network design under concurrent edge failures. Violating the relatively complete recourse assumption, their model allows second-stage infeasibility and leverages a two-level oracle to identify infeasible disruption scenarios and compute worst-case rerouting costs, solved via a tailored CCG algorithm.

However, our modeling methodology differs from the above related works on two-stage stochastic robust models in the following three aspects. (1) A demand function is embedded in our model. From a theoretical perspective, this function could be in an arbitrary form. If the function is nonlinear or non-closed form, it can break the model's linearity and make exact optimization intractable. Even when the function is linear, modeling demand as a price-dependent expression creates endogenous coupling between pricing and allocation decisions, which substantially increases the structural complexity of the model. (2) We incorporate resilience requirements as service constraints that must be satisfied under each disruption scenario. Instead of treating resilience as a performance indicator evaluated after optimization, we embed it directly into the feasible region of the second-stage allocation problem. This design forces pricing and capacity planning decisions to jointly ensure a minimum level of service reliability when the network is disrupted. (3) The proposed formulation integrates price-dependent uncertain demand, scenario-based supply disruptions, multimodal capacity planning, and resilience guarantees within a single optimization framework. To our knowledge, such a combination has not been jointly considered in prior research on pricing and allocation in logistics networks.

3. Problem description

Suppose a 4PL platform provides long-haul transport services to customers who need to transport a certain number of containers (in TEUs) from their origins to destinations. The platform does not own its transportation fleets, facilities and labor; thus, it needs to rent transportation capacity (in TEUs) from some third-party logistic (3PL) carriers such as shipping liners, train companies, and truck companies in advance, so as to fulfill the uncertain transportation demands released by the customers. In this problem, the platform aims to maximize its profit, which equals the revenue earned from the customers by fulfilling their demands minus the cost paid to the carriers for renting proper number of TEUs of transportation capacity. Fig. 1 provides the illustration of the multimodal network.

3.1. Multimodal transportation network under uncertain demand

The platform operates on a multimodal network consisting of a set of arcs C, indexed by c. As shown in Fig. 1(a), each arc represents a specific transportation mode (i.e., ship, rail, or truck) between two nodes. When multiple carriers serve the same connection with different rental prices, we treat them as distinct arcs. For each arc $c \in C$, the platform must decide the number of rented TEUs of transportation capacity, denoted by the integer decision variable α_c, at a unit renting cost w_c (USD/TEU). Customers' demands are categorized by a set of OD pairs N, indexed by n. For each OD pair $n \in N$, the platform quotes a unit price β_n (USD/TEU). The pricing decision β_n is bounded by a maximum acceptable price b_n, beyond which customers would resort to 3PL carriers directly. The nominal demand for OD pair n is characterized by a price-dependent function $f_n(\beta_n)$. To account for demand uncertainty, we introduce a demand scaled factor $\tilde{l}_{n\omega} = \bar{l}_n + z_{n\omega}\hat{l}_n$, where $\bar{l}_n$ is the nominal value and $\hat{l}_n$ is the maximum possible deviation. The uncertain term $z_{n\omega}$ is restricted by a polyhedral uncertainty set $\mathscr{U} = \left\{\tilde{l}_{n\omega} : \tilde{l}_{n\omega} = \bar{l}_n + z_{n\omega}\hat{l}_n,\ \forall n \in N,\ \omega \in \Omega,\ \boldsymbol{z} \in \mathscr{T}_\omega\right\}$, where $\mathscr{T}_\omega = \left\{\boldsymbol{z} | \sum_{n\in N} |z_{n\omega}| \le \Gamma_\omega, z_{n\omega} \in [-1, 1], \forall n \in N\right\}$. Here, Γ_ω serves as the budget of uncertainty, allowing decision-makers to adjust the conservatism level by limiting the number of demand factors that can deviate simultaneously from their nominal values.



Notes: In the typhoon scenario (b), ship arcs **Ningbo–Xiamen** and **Shanghai–Xiamen** are disrupted and become unavailable (remaining-capacity ratio = 0). In the earthquake scenario (c), the rail arc **Shanghai–Ningbo** is disrupted and its remaining capacity is reduced.

Fig. 1. The multimodal transportation network under different scenarios.

Operationally, Γ_ω acts as a tunable parameter capturing the overall level of mismatch between the estimated demand and the realized demand. A larger Γ_ω allows a larger total deviation across OD pairs (i.e., more OD pairs experiencing substantial forecast errors or many OD pairs experiencing moderate errors), leading to a more conservative plan; a smaller Γ_ω implies that the platform expects the estimation errors to be limited in total magnitude, resulting in a less conservative but potentially less robust plan. Consequently, the actual realized demand for OD pair n under scenario ω is expressed as $\widetilde{l}_{n\omega} f_n(\beta_n)$. Each OD pair can be served by a set of feasible transportation plans, denoted by R_n and indexed by r. Each plan $r \in R_n$ consists of a sequence of arcs, and we use a binary incidence parameter v_{rc} to indicate whether plan r traverses arc c($v_{rc} = 1$ if it uses arc c, and 0 otherwise). For example, in Fig. 1(a), the transportation demand of the OD pair Shanghai–Xiamen can be fulfilled by Plan A (Shanghai–Ningbo by rail and Ningbo–Xiamen by ship) or Plan B (Shanghai–Fuzhou by ship and Fuzhou–Xiamen by rail). After uncertainty is realized, the platform allocates transported volume across plans; the actual TEUs transported through plan r in scenario ω are denoted by the second-stage decision variable $\gamma_{r\omega}$.

3.2. Resilient network facing unusual emergencies

Some disruptive events such as floods, typhoons, and earthquakes may affect the transportation capacity in the network (Zhou et al. 2019, Wagenaar et al. 2020, Li et al. 2025). We represent each disruptive event as a disruption scenario and denote by Ω the set of all scenarios $\omega \in \Omega$, including the routine (no-disruption) scenario. Each scenario ω occurs with a known probability p_ω. Disruptions affect arc capacity through a scenario-dependent remaining-capacity ratio. Specifically, for each arc c and scenario ω, we define $g_{c\omega}$ as the percentage of remaining capacity on arc c under scenario ω. In the routine scenario, $g_{c\omega} = 1$, while in disruption scenarios $g_{c\omega} \in [0,1)$. Therefore, even if the platform rents α_c TEUs on arc c in advance, the usable capacity in scenario ω becomes $g_{c\omega}\alpha_c$. As shown in Fig. 1(b) and (c), these disruptions spatially impact the network. The platform's decision process is naturally formulated as a two-stage model embedded in a scenario-based framework. In the first stage, the platform determines the pre-schedule by choosing the rental decisions α_c and the pricing decisions β_n. In the second stage, when after scenario ω is realized and demand deviations are revealed within the uncertainty set, the platform determines the allocation decisions $\gamma_{r\omega}$ subject to the disrupted arc capacities $g_{c\omega}\alpha_c$. Based on this setting, the objective of the proposed model is to maximize the expected profit across all scenarios $\omega \in \Omega$ weighted by p_ω. It is noted that a robust optimization submodel that incorporates the uncertain demands of the OD pairs is embedded in each scenario of this study's model. Last but not the least, the resilience of the transportation network is also taken into account in the model's constraints. More specifically, according to the related works **(Byeon et al., 2020; Chen and Miller-Hooks, 2012; Wang et al., 2023), the network resilience is quantified as an indicator that equals the proportion of total transportation demand that the platform can actually fulfill under a disruption scenario (due to unusual emergency) and the worst-case demand fluctuation (due to uncertain demand). For each disruption scenario ω, the model requires this resilience indicator to be no smaller than a predefined threshold e_ω, thereby guaranteeing a minimum service capability under unusual emergencies and uncertain demand.

4. Stochastic programing model embedded with robust optimization

According to the above problem description, this section proposes a scenario-based stochastic programming model embedded with the robust optimization. The methodology of scenario-based stochastic programming is applied to handle the unusual emergencies that may occur in small probabilities, while the methodology of robust optimization is applied to handle the uncertain demands of OD pairs in each scenario. The rationale behind this hybrid modeling structure lies in the fundamentally different statistical characteristics of the two types of uncertainty. On the supply side, the available capacity of each transportation arc is affected by rare but high-impact disruption events (e.g., typhoons or floods), which can be naturally represented by a limited number of discrete scenarios with associated probabilities. Therefore, a scenario-based stochastic programming framework is suitable for describing such low-probability, high-severity supply disruptions (Sun et al. 2022, Yang et al. 2024). In contrast, on the demand side, the realized transportation demand for each OD pair exhibits continuous yet bounded variability due to market elasticity and data estimation errors. These variations cannot be easily characterized by explicit probability distributions but can be effectively captured by a polyhedral uncertainty set with an adjustable budget of uncertainty. Consequently, robust optimization provides a more appropriate tool to ensure stable performance against bounded demand fluctuations without assuming precise distributions (Wang and Qi 2020, Pu and Zhan 2021, Luo et al. 2025). By combining the scenario-based stochastic programming and the robust optimization, we formulate a two-stage MILP that maximizes the probability-weighted expected profit across disruption scenarios while evaluating operational revenue under worst-case demand deviations within F_ω in each scenario. As all key symbols have been defined within the context of the problem description in Section 3, we do not repeat them here to ensure a smooth exposition. Instead, a comprehensive notation table is available in Appendix for quick lookup.

$$[\mathscr{M}1]\ \text{Max} \sum_{\omega\in\Omega} p_\omega \min_{\boldsymbol{z}\in\mathscr{T}_\omega} \mathscr{C}_\omega(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{z}) - \sum_{c\in C} \boldsymbol{w}_c\alpha_c \tag{1-1}$$

subject to:

$$\beta_n \le b_n \ \forall n \in N \tag{1-2}$$

$$\alpha_c \ge 0 \ \forall c \in C \tag{1-3}$$

$$\beta_n \ge 0 \ \forall n \in N \tag{1-4}$$

Objective (1-1) maximizes the expected total profit, which equals the expected total revenue minus the total cost of renting transportation capacity on arcs. The $\sum_{\omega\in\Omega}p_\omega(\cdot)$ takes an expectation over disruption scenarios with known probabilities, reflecting a risk-neutral attitude toward supply disruptions. For each fixed scenario ω, the inner minimization $\min_{\boldsymbol{z}\in F_\omega}(\cdot)$ is applied only to the demand deviation vector $\boldsymbol{z}$, reflecting a distribution-free robust stance toward demand fluctuations when reliable demand distributions are unavailable. Therefore, the model should be viewed as maximizing the expected profit across scenarios, where the profit in each scenario is evaluated under the worst-case demand deviations permitted by F_ω. Constraints (1)-(2)–(1-4) define the domains of the first-stage decision variables $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, the values of which are transferred to the second-stage problem.

In the second-stage problem, the robust optimization handles the uncertain demands of OD pairs. In Objective (1-1), "$\min_{\boldsymbol{z}\in\mathcal{F}_\omega}\mathcal{Q}_\omega(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{z})$" evaluates the scenario ω operational revenue under the worst-case demand deviations within $\mathcal{F}_\omega$. More specifically, in scenario ω, $\mathcal{Q}_\omega(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{z})$ denotes the maximum revenue that can be earned by optimally deciding the actual number of containers (TEUs) transported via each transportation plan, i.e., the second-stage decision variable $\gamma_{r\omega}$. The second-stage model, whose objective is $\mathcal{Q}_\omega(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{z})$, is formulated as follows.

$$\mathcal{Q}_\omega(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{z}) = \text{Max}\sum_{n\in N}\beta_n\sum_{r\in R_n}\gamma_{r\omega} \tag{2-1}$$

subject to:

$$\sum_{r\in R_n}\gamma_{r\omega} \le (\bar{l}_n + z_{n\omega}\hat{l}_n)f_n(\beta_n)\ \forall n\in N \tag{2-2}$$

$$\sum_{n\in N}\sum_{r\in R_n}v_{rc}\gamma_{r\omega} \le g_{c\omega}\alpha_c\ \forall c\in C \tag{2-3}$$

$$\frac{\sum_{n\in N}\sum_{r\in R_n}\gamma_{r\omega}}{\sum_{n\in N}(\bar{l}_n + z_{n\omega}\hat{l}_n)f_n(\beta_n)} \ge e_\omega \tag{2-4}$$

$$\gamma_{r\omega} \ge 0\ \forall n\in N, r\in R_n. \tag{2-5}$$

Constraints (2)-(2) ensure that the number of actually transported containers on each OD pair does not exceed the OD pair's actual demand, which is dependent on the price $\boldsymbol{\beta}$ decided in the first stage. Constraints (2)-(3) guarantee that the total number of containers transported via each arc does not exceed the arc's available capacity, which is affected by the rented capacity $\boldsymbol{\alpha}$ decided in the first stage. Constraint (2)-(4) enforces the indicator of the network resilience in each scenario ω cannot be lower than a predefined threshold e_ω. Constraints (2)-(5) define the domain of the second-stage variable $\boldsymbol{\gamma}$. It is worth noting that the apparent nonlinearity in Constraints (2)-(4) does not affect model tractability. Since the price variables $\boldsymbol{\beta}$ are determined in the first stage and passed to the second stage as fixed parameters, the term $f_n(\beta_n)$ becomes a constant within the second-stage optimization. Consequently, the model does not impose restrictive structural assumptions on the demand function $f_n(\beta_n)$. Linear, piecewise linear, polynomial, or other empirically calibrated price-demand specifications can all be incorporated without affecting the validity or solvability of the model.

The max–min-max objective structure provides a layered evaluation of uncertainty and mirrors the platform's planning-and-response logic. The outer maximization captures the platform's ex-ante pricing and capacity planning decisions $(\boldsymbol{\alpha},\boldsymbol{\beta})$. Scenario uncertainty $\omega\in\Omega$ represents supply-side disruptions with known probabilities and is aggregated through the expectation operator $\sum_{\omega\in\Omega}p_\omega(\cdot)$. Conditional on each scenario ω, the inner minimization over $\boldsymbol{z}\in F_\omega$ evaluates revenue under worst-case demand deviations, and the inner maximization defining $Q_\omega(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{z})$ represents the platform's best operational response $\boldsymbol{\gamma}$ under the realized capacities and demand realization. This nested formulation is not a mathematically imposed construction. Instead, it reflects the platform's decision logic, separating ex-ante planning, uncertainty evaluation, and operational recourse. Consequently, the three-level structure enhances the practical relevance of the model, because each layer corresponds to a real and commonly observed decision stage in platform-based logistics systems operating under uncertainty. Although the model evaluates worst-case demand realizations, this does not make the formulation overly conservative. The uncertainty budget Γ_ω explicitly controls the severity of adverse deviations considered, allowing the decision maker to calibrate the desired robustness level. Moreover, the inner maximization ensures that, even under unfavorable conditions, the platform is assessed based on its best operational response, thereby preventing excessive pessimism.

5. Algorithm

CCG introduced by Zeng and Zhao (2013) is one of the two mainstream decomposition frameworks commonly used for solving max–min–max or min–max–min robust optimization models, with the other being Benders decomposition (Matthews et al. 2019, Wang et al. 2024, Luo et al. 2025). Existing studies have shown that CCG often achieves better computational performance than Benders decomposition in most cases (Zeng and Zhao, 2013, An et al. 2014, Wang and Qi 2020, Wang et al. 2025). The algorithm alternates between solving a master problem to obtain first-stage solutions and solving a subproblem to obtain second-stage solutions. A key assumption in the classical CCG framework is the relatively complete recourse, which ensures that the subproblem remains feasible under all uncertainty realizations for any given first-stage decision. However, in our multimodal network design problem, this assumption does not hold. Because arc capacities may be significantly reduced or disrupted under certain scenarios, the resilience constraints can render the second-stage allocation problem infeasible for some first-stage capacity decisions. When this occurs, a

classical CCG algorithm that does not include feasibility verification would encounter an infeasible subproblem, terminate prematurely, and be unable to generate valid cuts or feasible solutions. Therefore, the classical CCG procedure cannot be applied directly to our model. To address this issue, we generalize the CCG framework by introducing a dedicated feasibility subproblem. The purpose of this subproblem is to identify infeasible recourse decisions and generate feasibility cuts that remove such infeasible regions from the first-stage decision space. This component is essential for restoring the solvability of the algorithm, because it ensures that the CCG procedure can continue even when the relatively complete recourse assumption is violated. With this extension, the CCG framework becomes capable of handling incomplete recourse and converging reliably for the resilient intermodal network design problem.

Despite this enhancement, CCG still faces common scalability challenges inherent in exact algorithms, particularly when the number of disruption scenarios becomes large and the second-stage evaluation must be repeatedly solved across iterations (Yang and Zhen 2025). To alleviate this computational bottleneck, we further develop a CPU-GPU hybrid heuristic that accelerates second-stage evaluations via parallelization on GPUs, enabling the framework to handle substantially larger instances within limited time. The two algorithms are complementary rather than redundant. The extended CCG procedure provides an exact benchmark for validating the mathematical formulation and quantifying the optimality gap of the heuristic on small-to-medium instances. The CPU-GPU heuristic is employed to tackle large-scale problems where the iterative MILP solves of CCG become computationally prohibitive.

5.1. Model transformation

To support the extended CCG framework, we reformulate the original second-stage problem into two subproblems: a feasibility subproblem and an optimality subproblem. The feasibility subproblem is designed to detect whether a given first-stage solution leads to infeasibility under any uncertainty realization. When the solution is feasible, the optimality subproblem identifies the worst-case scenario that degrades the objective value. The following subsections describe the construction and transformation of these two subproblems in detail.

5.1.1. Feasibility subproblem

To enable feasibility checking, we reformulate the original second-stage problem by introducing non-negative slack variables $\zeta_{r\omega}$ into the resilience constraints (i.e., Constraints (2)-(4) are reformulated as Constraints (3)-(2)) and modifying the objective function as the negative sum of slack variables. These slack variables ensure that the subproblem remains feasible under all realizations of uncertainty, thereby satisfying the relatively complete recourse assumption. By maximizing the negative sum of slack variables, we aim to identify the minimal level of violation needed to restore feasibility. If the optimal value is zero, the original subproblem is feasible under the given first-stage solution; otherwise, the presence of positive slack variables indicates infeasibility. The modified inner maximization problem is as follows:

$$\mathscr{R}_\omega(\overline{\boldsymbol{\alpha}},\overline{\boldsymbol{\beta}},\overline{\boldsymbol{z}}) = \text{Max} - \sum_{n\in N}\sum_{r\in R_n}\zeta_{r\omega} \tag{3-1}$$

subject to: Constraints (2)-(2)–(2-3), and (2-5);

$$\frac{\sum_{n\in N}\sum_{r\in R_n}(\gamma_{r\omega}+\zeta_{r\omega})}{\sum_{n\in N}(\bar{l}_n+\overline{z}_{n\omega}\hat{l}_n)f_n(\bar{\beta}_n)} \geq e_\omega \tag{3-2}$$

$$\zeta_{r\omega} \geq 0. \tag{3-3}$$

Then, the part "$\min_{\boldsymbol{z}\in\mathscr{T}_\omega}\mathscr{C}_\omega(\overline{\boldsymbol{\alpha}},\overline{\boldsymbol{\beta}},\boldsymbol{z})$" in Objective (1-1) is reformulated as "$\min_{\boldsymbol{z}\in\mathscr{T}_\omega}\mathscr{R}_\omega(\overline{\boldsymbol{\alpha}},\overline{\boldsymbol{\beta}},\boldsymbol{z})$"; here, the $\overline{\boldsymbol{\alpha}}$ and $\overline{\boldsymbol{\beta}}$ are the values of the first-stage solution. The "$\min_{\boldsymbol{z}\in\mathscr{T}_\omega}\mathscr{R}_\omega(\overline{\boldsymbol{\alpha}},\overline{\boldsymbol{\beta}},\boldsymbol{z})$" is a minimization problem, which includes the maximization problem $\mathscr{R}_\omega(\bullet)$, i.e., "Max $-\sum_{n\in N}\sum_{r\in R_n}\zeta_{r\omega}$". Thus, we should first transform the maximization problem $\mathscr{R}_\omega(\bullet)$ to an equivalent minimization problem; then the above part could be solvable. More specifically, given a specific value of $\overline{\boldsymbol{z}}$ realized in the uncertainty set, we need to equivalently transfer the maximization problem $\mathscr{R}_\omega(\overline{\boldsymbol{\alpha}},\overline{\boldsymbol{\beta}},\overline{\boldsymbol{z}})$ to a minimization problem. The transformation process is explained as follows. The dual variables for Constraints (2)-(2), (2-3), and (3-2) are denoted by $\varepsilon^1_{n\omega}$, $\varepsilon^2_{c\omega}$, and ε^3_ω, respectively. According to the dual theory, the dual problem of $\mathscr{R}_\omega(\overline{\boldsymbol{\alpha}},\overline{\boldsymbol{\beta}},\overline{\boldsymbol{z}})$ is formulated as:

$$\text{Min}\sum_{n\in N}\varepsilon^1_{n\omega}(\bar{l}_n+\overline{z}_{n\omega}\hat{l}_n)f_n(\bar{\beta}_n)+\sum_{c\in C}\varepsilon^2_{c\omega}g_{c\omega}\overline{\alpha}_c-\varepsilon^3_\omega e_\omega\sum_{n\in N}(\bar{l}_n+\overline{z}_{n\omega}\hat{l}_n)f_n(\bar{\beta}_n) \tag{4-1}$$

subject to:

$$\varepsilon^1_{n\omega}+\sum_{c\in C}\varepsilon^2_{c\omega}v_{rc}-\varepsilon^3_\omega \geq 0 \ \forall n\in N, r\in R_n \tag{4-2}$$

$$1-\varepsilon^3_\omega \geq 0 \ \forall n\in N, r\in R_n \tag{4-3}$$

$$\varepsilon^1_{n\omega} \geq 0 \ \forall n\in N \tag{4-4}$$

$$\varepsilon^2_{c\omega} \geq 0 \ \forall c\in C \tag{4-5}$$

$$\varepsilon_{\omega}^{3} \geq 0 \tag{4-6}$$

By applying strong duality, the "$\min_{\boldsymbol{z}\in\mathcal{T}_{\omega}} \mathscr{R}_{\omega}(\overline{\boldsymbol{\alpha}}, \overline{\boldsymbol{\beta}}, \boldsymbol{z})$" is transformed from a min–max structure into an equivalent min-min problem. Then we further reduce this min-min problem to a single-level minimization problem, in which the vector $\boldsymbol{z}$ is the decision variable. However, this introduces bilinear terms such as $\varepsilon_{n\omega}^{1} z_{n\omega}$ and $\varepsilon_{\omega}^{3} z_{n\omega}$ in Objective (4-1), which results in nonlinearity. To address this issue, we apply the reformulation technique proposed by Zeng and Zhao (2013), which enables linearization of such bilinear terms and facilitates solving the problem as a single-level MILP. The reformulated model for the "$\min_{\boldsymbol{z}\in\mathcal{T}_{\omega}} \mathscr{R}_{\omega}(\overline{\boldsymbol{\alpha}}, \overline{\boldsymbol{\beta}}, \boldsymbol{z})$" is shown as follows.

$$[\mathscr{M}2]\min -\sum_{n\in N}\sum_{r\in R_n}\zeta_{r\omega} \tag{5-1}$$

subject to: Constraints (2)-(2)–(2-3), (2-5), (3-2)–(3-3), and (4-2)–(4-6);

$$\varepsilon_{n\omega}^{1}\left[(\bar{l}_n + z_{n\omega}\widehat{l}_n) f_n(\overline{\beta}_n) - \sum_{r\in R_n}\gamma_{r\omega}\right] = 0 \ \forall n \in N \tag{5-2}$$

$$\varepsilon_{c\omega}^{2}\left[g_{c\omega}\overline{\alpha}_c - \sum_{n\in N}\sum_{r\in R_n}\nu_{rc}\gamma_{r\omega}\right] = 0 \ \forall c \in C \tag{5-3}$$

$$\varepsilon_{\omega}^{3}\left[\sum_{n\in N}\sum_{r\in R_n}(\gamma_{r\omega} + \zeta_{r\omega}) - e_{\omega}\sum_{n\in N}(\bar{l}_n + z_{n\omega}\widehat{l}_n) f_n(\overline{\beta}_n)\right] = 0 \tag{5-4}$$

$$\gamma_{r\omega}\left(\varepsilon_{n\omega}^{1} + \sum_{c\in C}\varepsilon_{c\omega}^{2}\nu_{rc} - \varepsilon_{\omega}^{3}\right) = 0 \ \forall n \in N, r \in R_n \tag{5-5}$$

$$\zeta_{r\omega}\left(1 - \varepsilon_{\omega}^{3}\right) = 0 \ \forall n \in N, r \in R_n \tag{5-6}$$

$$\sum_{n\in N}|z_{n\omega}| \leq \Gamma_{\omega} \tag{5-7}$$

$$z_{n\omega} \in [-1, 1] \forall n \in N. \tag{5-8}$$

Constraints (5)-(2)–(5-6) are complementary slackness conditions. By using the big-M method, Constraints (5)-(2), (5-3), (5-4), (5-5), and (5-6) can be linearized by introducing auxiliary binary variables $\iota_{n\omega}^{1}$, $\iota_{c\omega}^{2}$, $[MISDI]_{\omega}^{3}$, $\iota_{r\omega}^{4}$, and $\iota_{r\omega}^{5}$ respectively. In particular, Constraints (5)-(2)–(5-6) are linearized as follows.

$$\varepsilon_{n\omega}^{1} \leq M\iota_{n\omega}^{1} \ \forall n \in N \tag{5-9}$$

$$(\bar{l}_n + z_{n\omega}\widehat{l}_n) f_n(\overline{\beta}_n) - \sum_{r\in R_n}\gamma_{r\omega} \leq M\left(1 - \iota_{n\omega}^{1}\right) \ \forall n \in N \tag{5-10}$$

$$\varepsilon_{c\omega}^{2} \leq M\iota_{c\omega}^{2} \forall c \in C \tag{5-11}$$

$$g_{c\omega}\overline{\alpha}_c - \sum_{n\in N}\sum_{r\in R_n}\nu_{rc}\gamma_{r\omega} \leq M\left(1 - \iota_{c\omega}^{2}\right) \ \forall c \in C \tag{5-12}$$

$$\varepsilon_{\omega}^{3} \leq M[MISDI]_{\omega}^{3} \tag{5-13}$$

$$\sum_{n\in N}\sum_{r\in R_n}(\gamma_{r\omega} + \zeta_{r\omega}) - e_{\omega}\sum_{n\in N}(\bar{l}_n + z_{n\omega}\widehat{l}_n) f_n(\overline{\beta}_n) \leq M\left(1 - [MISDI]_{\omega}^{3}\right) \tag{5-14}$$

$$\gamma_{r\omega} \leq M\iota_{r\omega}^{4} \ \forall n \in N, r \in R_n \tag{5-15}$$

$$\varepsilon_{n\omega}^{1} + \sum_{c\in C}\varepsilon_{c\omega}^{2}\nu_{rc} - \varepsilon_{\omega}^{3} \leq M\left(1 - \iota_{r\omega}^{4}\right) \ \forall n \in N, r \in R_n \tag{5-16}$$

$$\zeta_{r\omega} \leq M\iota_{r\omega}^{5} \ \forall n \in N, r \in R_n \tag{5-17}$$

$$1 - \varepsilon_{\omega}^{3} \leq M\left(1 - \iota_{r\omega}^{5}\right) \ \forall n \in N, r \in R_n \tag{5-18}$$

If the optimal value of model $\mathscr{M}2$ is nonzero, then the first-stage solution $\overline{\boldsymbol{\alpha}}$ and $\overline{\boldsymbol{\beta}}$ are robust infeasible, and the optimal vector $\boldsymbol{z}$ stemming from this problem can be used to further restrict the decision variables in the master problem. The value of vector $\boldsymbol{z}$ is stored

and explicitly used to generate new constraints in the master problem. However, if the optimal value of model $\mathscr{M}2$ is zero, then the first-stage solution $\overline{\boldsymbol{\alpha}}$ and $\overline{\boldsymbol{\beta}}$ are robust feasible, and the optimality subproblem must be solved to determine actual worst-case profit of these decision variables of first stage problem.

5.1.2. Optimality subproblem

Once the feasibility of the current first-stage solution $(\overline{\boldsymbol{\alpha}}, \overline{\boldsymbol{\beta}})$ is confirmed, the optimality subproblem is constructed and used to identify the worst-case disruption scenario under this solution. Similar to the feasibility subproblem, we reformulate this problem by first deriving the dual of the second-stage maximization problem and then apply the KKT conditions to obtain a single-level MILP that can be solved efficiently. The transformation process is explained as follows.

$$\mathscr{C}_{\omega}(\overline{\boldsymbol{\alpha}}, \overline{\boldsymbol{\beta}}, \overline{\boldsymbol{z}}) = \text{Max} \sum_{n \in N} \overline{\beta}_n \sum_{r \in R_n} \gamma_{r\omega} \tag{6-1}$$

subject to: Constraints (2)-(2)–(2-5).

The dual variables for Constraints (2)-(2), (2-3), and (2-4) are denoted by $\tau^1_{n\omega}$, $\tau^2_{c\omega}$, and τ^3_{ω}, respectively. According to the dual theory, the dual problem of $\mathscr{C}_{\omega}(\overline{\boldsymbol{\alpha}}, \overline{\boldsymbol{\beta}}, \overline{\boldsymbol{z}})$ is formulated as:

$$\text{Min} \sum_{n \in N} \tau^1_{n\omega} (\bar{l}_n + \overline{z}_{n\omega} \widehat{l}_n) f_n(\overline{\beta}_n) + \sum_{c \in C} \tau^2_{c\omega} g_{c\omega} \overline{\alpha}_c - \tau^3_{\omega} e_{\omega} \sum_{n \in N} (\bar{l}_n + \overline{z}_{n\omega} \widehat{l}_n) f_n(\overline{\beta}_n) \tag{7-1}$$

subject to:

$$-\overline{\beta}_n + \tau^1_{n\omega} + \sum_{c \in C} \tau^2_{c\omega} v_{rc} - \tau^3_{\omega} \geq 0 \ \forall n \in N, r \in R_n \tag{7-2}$$

$$\tau^1_{n\omega} \geq 0 \ \forall n \in N \tag{7-3}$$

$$\tau^2_{c\omega} \geq 0 \ \forall c \in C \tag{7-4}$$

$$\tau^3_{\omega} \geq 0 \tag{7-5}$$

The reformulated model for the original "$\min_{\boldsymbol{z} \in \mathscr{T}_{\omega}} \mathscr{C}_{\omega}(\overline{\boldsymbol{\alpha}}, \overline{\boldsymbol{\beta}}, \boldsymbol{z})$", which initially takes the form of a min–max problem, is converted into a single-level minimization model $\mathscr{M}3$ through duality transformation and is shown as follows.

$[\mathscr{M}3] \min \sum_{n \in N} \overline{\beta}_n \sum_{r \in R_n} \gamma_{r\omega}$ (8-1)

subject to: Constraints (2)-(2)–(2-5), and (4-2)–(4-5);

$$\tau^1_{n\omega} \left[(\bar{l}_n + z_{n\omega} \widehat{l}_n) f_n(\overline{\beta}_n) - \sum_{r \in R_n} \gamma_{r\omega} \right] = 0 \ \forall n \in N \tag{8-2}$$

$$\tau^2_{c\omega} \left[g_{c\omega} \overline{\alpha}_c - \sum_{n \in N} \sum_{r \in R_n} v_{rc} \gamma_{r\omega} \right] = 0 \ \forall c \in C \tag{8-3}$$

$$\tau^3_{\omega} \left[\sum_{n \in N} \sum_{r \in R_n} \gamma_{r\omega} - e_{\omega} \sum_{n \in N} (\bar{l}_n + z_{n\omega} \widehat{l}_n) f_n(\overline{\beta}_n) \right] = 0 \tag{8-4}$$

$$\gamma_{r\omega} \left(-\overline{\beta}_n + \tau^1_{n\omega} + \sum_{c \in C} \tau^2_{c\omega} v_{rc} - \tau^3_{\omega} \right) = 0 \ \forall n \in N, r \in R_n \tag{8-5}$$

$$\sum_{n \in N} |z_{n\omega}| \leq \Gamma_{\omega} \tag{8-6}$$

$$z_{n\omega} \in [-1, 1] \ \forall n \in N. \tag{8-7}$$

Constraints (8)-(2)–(8-5) are complementary slackness conditions. By using the big-M method, Constraints (8)-(2), (8-3), (8-4), and (8-5) can be linearized by introducing auxiliary binary variables $\sigma^1_{n\omega}$, $\sigma^2_{c\omega}$, σ^3_{ω}, and $\sigma^4_{r\omega}$, respectively. In particular, Constraints (8)-(2)–(8-5) are linearized as follows.

$$\tau^1_{n\omega} \leq M \sigma^1_{n\omega} \forall n \in N \tag{8-8}$$

$$(\bar{l}_n + z_{n\omega} \widehat{l}_n) f_n(\overline{\beta}_n) - \sum_{r \in R_n} \gamma_{r\omega} \leq M\left(1 - \sigma^1_{n\omega}\right) \ \forall n \in N \tag{8-9}$$

$$\tau^2_{c\omega} \leq M \sigma^2_{c\omega} \ \forall c \in C \tag{8-10}$$

$$g_{c\omega}\overline{\alpha}_c - \sum_{n\in N}\sum_{r\in R_n} v_{rc}\gamma_{r\omega} \le M\left(1-\sigma^2_{c\omega}\right)\ \forall c\in C \tag{8-11}$$

$$\tau^3_\omega \le M\sigma^3_\omega \tag{8-12}$$

$$\sum_{n\in N}\sum_{r\in R_n}\gamma_{r\omega} - e_\omega\sum_{n\in N}(\bar{l}_n + z_{n\omega}\hat{l}_n)f_n(\overline{\beta}_n) \le M\left(1-\sigma^3_\omega\right) \tag{8-13}$$

$$\gamma_{r\omega} \le M\sigma^4_{r\omega}\ \forall n\in N, r\in R_n \tag{8-14}$$

$$-\overline{\beta}_n + \tau^1_{n\omega} + \sum_{c\in C}\tau^2_{c\omega}v_{rc} - \tau^3_\omega \le M\left(1-\sigma^4_{r\omega}\right)\ \forall n\in N, r\in R_n \tag{8-15}$$

This optimality subproblem enables the identification of the worst-case disruption scenario under the current decision, and the solution is used to generate optimality cuts in the master problem.

5.2. Column-and-constraint generation

Based on the above transformation of the second-stage problem, we use the CCG to solve the original master problem $\mathscr{M}1$. The CCG is a multi-iteration procedure. Let the index of the iterations is defined as k. In each iteration, the master problem is updated by adding new constraints associated with the new disruption scenario. Let $\Theta \subseteq \{0,1,2,\ldots,k\}$ denote the subset of iterations, that the identified disruption scenario leads to an optimality cut (i.e., a constraint involving η). That is, k tracks all iterations, while Θ only collects those iterations contributing optimality information to the master problem. In the yth iteration of the CCG, we solve the following model $\mathscr{M}4$, which is formulated as follows.

$[\mathscr{M}4]\min \eta - \sum_{c\in C} w_c\alpha_c$ (9-1)

subject to: Constraints (1)-(2)–(1-4);

$$\eta \le \sum_{\omega\in\Omega} p_\omega \sum_{n\in N}\beta_n \sum_{r\in R_n}\gamma^k_{r\omega}\ \forall k\in\Theta \tag{9-2}$$

$$\sum_{r\in R_n}\gamma^k_{r\omega} \le \left(\bar{l}_n + z^k_{n\omega}\hat{l}_n\right)f_n(\beta_n)\ \forall\omega\in\Omega, n\in N, k=0,1,\cdots,y \tag{9-3}$$

$$\sum_{n\in N}\sum_{r\in R_n} v_{rc}\gamma^k_{r\omega} \le g_{c\omega}\alpha_c\ \forall\omega\in\Omega, c\in C, k=0,1,\cdots,y \tag{9-4}$$

$$\frac{\sum_{n\in N}\sum_{r\in R_n}\gamma^k_{r\omega}}{\sum_{n\in N}\left(\bar{l}_n + z^k_{n\omega}\hat{l}_n\right)f_n(\beta_n)} \ge e_\omega\ \forall\omega\in\Omega, k=0,1,\cdots,y \tag{9-5}$$

$$\eta \ge 0. \tag{9-6}$$

$$\gamma^k_{r\omega} \ge 0\ \forall\omega\in\Omega, n\in N, r\in R_n, k=0,1,\cdots,y \tag{9-10}$$

In the above model $\mathscr{M}4$, the values of $\gamma^k_{r\omega}$ and $z^k_{n\omega}$ are parameters, which were obtained by solving the model in the kth iteration. Based on these values, Constraints (9)-(2)–(9-5), which are originally in the second stage, are added (merged) into the above master problem. The CCG algorithm progressively enriches the master problem by incorporating information from the second-stage evaluation. When the feasibility subproblem identifies a disruption scenario under which the current first-stage solution is infeasible, a feasibility cut is generated and appended to the master problem to restore feasibility. When the optimality subproblem identifies the worst-case scenario, an optimality cut is added to tighten the lower bound. Through this iterative process, the master problem increasingly approximates the true recourse function until convergence. The upper and lower bounds of the original problem are denoted by UB and LB, respectively. The relative optimality gap is defined as $GAP = UB - LB$. The pseudocode of CCG is as follows.

Algorithm: Column-and-constraint generation

Input: Parameters of all arcs and OD-pairs, network resilience indicators of all scenarios,*optimalitytolerance*

Output: Best solution $(\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*, \boldsymbol{\gamma}^*)$

1: Set $y\leftarrow 0$, $UB\leftarrow +\infty$, $LB\leftarrow -\infty$, $\Theta\leftarrow\varnothing$, and $GAP\leftarrow +\infty$

2: Set the master problem $\mathscr{M}4$ and solve it to obtain the initial first-stage solution $(\boldsymbol{\alpha}^{(y)}, \boldsymbol{\beta}^{(y)})$

3: **While** $GAP \ge optimalitytolerance$ **do**

4: Solve the master model $\mathscr{M}4$ and obtain $(\boldsymbol{\alpha}^*_{y+1}, \boldsymbol{\beta}^*_{y+1}, \eta^*_{y+1}, \boldsymbol{\gamma}^{1^*}, \cdots, \boldsymbol{\gamma}^{y^*})$

5: Update the upper bound $UB\leftarrow\min\left\{UB, \eta^*_{y+1} - \boldsymbol{w}^T\boldsymbol{\alpha}^*_{y+1}\right\}$

6: Solve feasibility subproblem $\mathscr{M}2$ to evaluate $\left(\boldsymbol{\alpha}^*_{y+1}, \boldsymbol{\beta}^*_{y+1}\right)$

(*continued on next page*)

(*continued*)

7: **If** subproblem $\mathscr{M}2$ is infeasible **then**
8: Identify infeasible scenario $\boldsymbol{z}^{y+1}$
9: Add the feasibility cut (Constraints (9)-(3)–(9-5)) to master problem $\mathscr{M}4$
10: Set $y \leftarrow y + 1$
11: **Continue**
12: **End if**
13: Solve optimality subproblem $\mathscr{M}3$ to compute the worst-case scenario $\boldsymbol{z}^{y+1}$
14: Update the $LB \leftarrow \max\left\{LB, \sum_{\omega\in\Omega} p_\omega \min_{\boldsymbol{z}\in\mathscr{Z}_\omega} \mathscr{C}_\omega\left(\boldsymbol{\alpha}^*_{y+1}, \boldsymbol{\beta}^*_{y+1}, \boldsymbol{z}\right) - \boldsymbol{w}^T\boldsymbol{\alpha}^*_{y+1}\right\}$
15: Compute the optimality gap $GAP \leftarrow UB - LB$
16: **If** $GAP <$ *optimalitytolerance* **then break**
17: Add the optimality cuts (Constraints (9)-(2)–(9-5)) from $\boldsymbol{z}^{y+1}$
18: Set $y \leftarrow y + 1$, $\Theta \leftarrow \Theta \cup \{y\}$
19: **End while**
20: Return $(\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*, \boldsymbol{\gamma}^*)$

5.3. CPU-GPU hybrid heuristic algorithm

To overcome the computational burden faced by the CCG algorithm when the problem scale becomes large, we design a CPU-GPU hybrid heuristic algorithm that can efficiently produce high-quality solutions within limited time. This heuristic is not intended to replace CCG as a benchmark method; instead, it is designed for scalability, while its solution quality is assessed against CCG benchmarks on tractable instances. This design is inspired by recent advances in heterogeneous computing for large-scale optimization, where CPU-GPU hybrid architectures have proven effective in accelerating decomposition-based algorithms (Shi et al. 2026). The method uses a decomposition-based framework: the first-stage procedure, executed on the CPU, updates the strategic decision variables, i.e., capacity planning decisions α_c and pricing decisions β_n; the second-stage procedure, executed on the GPU, evaluates the flow assignment across all disruption realizations in parallel. In this framework, the CPU coordinates the global search and decision adjustments, while the GPU significantly accelerates the evaluation of the second-stage problem via parallelization. The overall procedure is outlined below.

Step 1. The CPU generates an initial solution $(\boldsymbol{\alpha}, \boldsymbol{\beta})$ that provides a reasonable and structurally balanced starting point for the subsequent search process. Arcs are first ranked by their occurrence frequency in the candidate transportation plans. Based on this ranking, arc capacities are initialized by assigning relatively larger values to arcs with higher coverage ratios and smaller values to arcs with lower coverage ratios, ensuring that the initial solution reflects the structural importance of different arcs. The initial value of β_n are drawn uniformly from their feasible intervals, thereby preventing overly biased initial price configurations.

Step 2. For each scenario ω, the CPU constructs the worst-case disturbance vector $z_{n\omega}$ subject to the uncertainty budget Γ_ω, and computes the demand $f_n(\beta_n)$. All required parameters for every scenario are then transferred to the GPU.

Step 3. On the GPU, each disruption scenario ω is assigned to an independent compute block. At the beginning of the computation, the block loads its scenario-specific parameters, including the arc availability ratios $g_{c\omega}$, the disturbance vector $z_{n\omega}$, and the transportation plan–arc incidence data v_{rc}, from global memory into shared memory. This initialization step ensures that all subsequent operations within the block can access data with minimal memory latency.

Step 4. The compute threads within each block collaboratively calculate the scenario-specific effective arc capacities $\mathrm{cap}_c = \alpha_c \cdot g_{c\omega}$, and the disturbed scenario demand $d_n = (l_n + z_{n\omega}\widehat{l}_n) \cdot f_n(\beta_n)$. The resulting values are stored in shared memory so that all subsequent operations in the block can access them efficiently.

Step 5. Each thread scans a subset of candidate transportation plans and computes its bottleneck capacity $u_r = \min_{c:v_{rc}>0} \mathrm{cap}_c v_{rc}$. All results are written to shared memory.

Step 6. To maintain consistency with the reference greedy assignment rule, a designated control thread within the block performs a stable descending sort of the transportation plans based on u_r. The resulting order uniquely determines the flow assignment sequence.

Step 7. Given the sorted order, transportation plan assignment proceeds iteratively: (i) Thread 0 computes the loading quantity $f_r = \min(d_n, u_r)$, and updates the remaining demand $d_n \leftarrow d_n - f_r$; (ii) All threads update arc capacities in parallel $\mathrm{cap}_c \leftarrow (\mathrm{cap}_c - f_r v_{rc})$; (iii) Updates the scenario revenue as $Q_\omega \leftarrow Q_\omega + f_r\beta_n$; (iv) Synchronization is performed after each update to maintain shared-memory consistency.

Step 8. A designated control thread within the block computes the scenario-level service rate; if it falls below the resilience threshold e_ω, the scenario revenue Q_ω is replaced by a penalty constant.

Step 9. A designated control thread within the block writes the scenario revenue Q_ω, the final arc capacities, and the remaining demand back to global memory, enabling the CPU to aggregate the multi-scenario results.

Step 10. The CPU aggregates scenario outcomes and adjusts the decision variables: (i) Arc capacities with consistently positive residual capacity across scenarios are reduced to avoid over-allocation; (ii) OD prices corresponding to systematically unmet demand are increased to reallocate capacity; (iii) A single-variable neighborhood search is applied by slightly perturbing each α_c or β_n and re-evaluating the inner problem, accepting the change only if improvement is observed; (iv) If no improvement occurs across multiple iterations, a random perturbation mechanism is triggered to escape local optimal.

Step 11. The algorithm terminates when the maximum number of iterations is reached and the best solution obtained is returned.

6. Computational experiments

We conduct a series of comprehensive computational experiments to evaluate the applicability and effectiveness of the proposed models and algorithms. Section 6.1 introduces a real-world case as the basis for constructing the experimental instances. Next, we test the performance of the scenario-based stochastic programming model embedded with the robust optimization in Section 6.2. Section 6.3 evaluates the computational efficiency of the proposed CPU-GPU hybrid heuristic by comparing its performance with the CCG algorithm, highlighting the scalability improvements achieved through GPU-based parallelization. In addition, sensitivity analyses are conducted to examine the effects of key uncertain parameters, and the corresponding managerial insights are discussed in Section 6.4. All experiments are performed on a workstation equipped with an Intel Xeon E5-2643 v3 CPU @ 3.40 GHz and 128 GB of RAM. The algorithms are implemented in Python (version 3.11.5), and all optimization models are solved using GUROBI 12.0.

6.1. Real case

The experimental data are obtained from a third-party logistics platform that operates a real-world multimodal transportation network in Southeast China. The company provides container transportation services among four cities: Shanghai, Ningbo, Fuzhou, and Xiamen. There are three kinds of transportation vehicles (i.e., trains, ships, and trucks) in this network. The network includes 20 directed arcs. The structure of the multimodal transportation network is shown in Fig. 2. Based on the four cities, we consider ten OD pairs. To model the demand function for each OD pair, we analyze historical operational data from the case company, which contain matched price–volume observations. Exploratory data analysis shows a clear linear downward pattern between quoted prices and realized booking quantities, motivating the use of a linear specification. Accordingly, we estimate $f_n(x) = m_n x + u_n$ for each OD pair via regression, and the fitted models exhibit strong explanatory power (R-squared values exceeding 0.9 and p-values below 0.01 in F-tests), confirming that the linear form is appropriate for the empirical case. Similar linear demand formulations have been employed in the existing literature (Liu and Yang 2015, Wang et al. 2015, Wang and Meng 2021). For each OD pair n, the demand function satisfies $f_n(b_n) = 0$, which implies that $m_n < 0$ and $u_n > 0$. The unit transportation and arc rental costs per container, ranging from \$100 to \$500 and \$50 to \$400, respectively, are based on actual pricing data from the case company. For each OD pair, a set of feasible transportation plans is generated in advance based on the available arcs and their modal connections in the network.

In this study, typhoons are considered the primary source of disruption. According to the official classification by the China Meteorological Administration (CMA), tropical cyclones are divided into six intensity levels based on their maximum sustained wind speeds: tropical depression (TD), tropical storm (TS), severe tropical storm (STS), typhoon (TY), severe typhoon (STY), and super typhoon (SuperTY) (CMA, 2009). In this study, we focus on the five higher levels (TS to SuperTY), which are more likely to cause significant disruptions to transportation systems. Based on the recorded typhoon events in China from 2020 to 2024, the empirical probabilities of occurrence for TS, STS, TY, STY, and SuperTY events are estimated to be 0.06, 0.03, 0.005, 0.003, and 0.002, respectively. For each level, we define a set of distinct disruption scenarios that represent variations in the affected location and severity. The sum of the disruption scenario probabilities p_ω under each level equals the empirical occurrence probability of that typhoon level. For example, suppose we define three disruption scenarios under the STS category: ω_6, ω_7, and ω_8, with corresponding

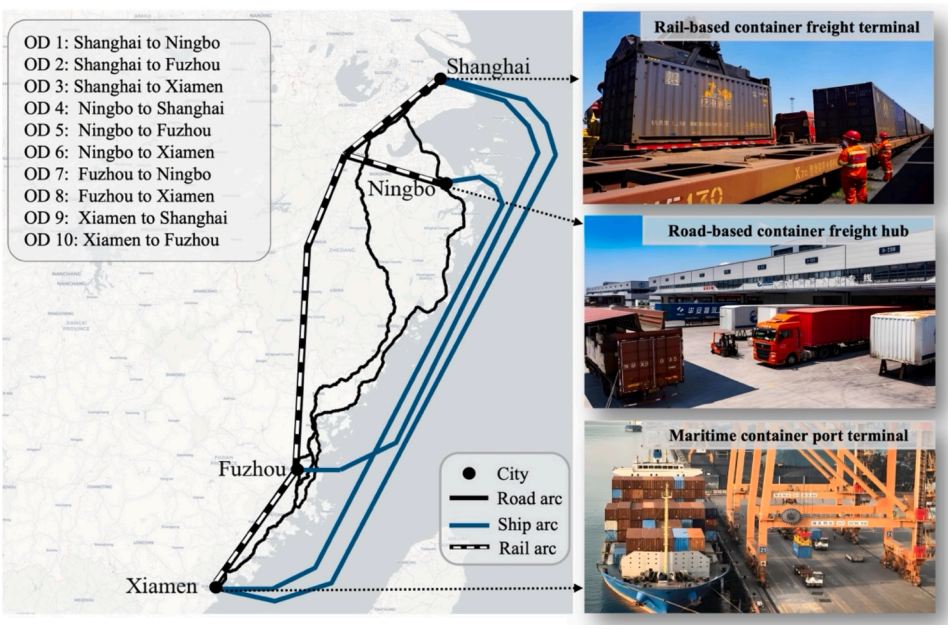


Fig. 2. The distribution of the multimodal transportation network.

probabilities $p_{\omega_6} = 0.015$, $p_{\omega_7} = 0.01$, $p_{\omega_8} = 0.005$. These disruption scenarios involve different subsets of arcs being disrupted by moderate typhoon intensity, and their total probability $p_{\omega_6} + p_{\omega_7} + p_{\omega_8} = 0.03$ aligns with the empirical frequency of STS level typhoons. The percentage of remaining capacity of each disruption scenario is calculated by considering the traffic control rules associated with each typhoon level and the proportion of affected arcs. For example, under a TS-level typhoon, highway speed limits for trucks are typically reduced from 100 km/h to 80 km/h, while rail and maritime operations are generally unaffected. If five out of the 20 arcs in the network are affected by such speed restrictions, the remaining capacity ratio is estimated as $1 - (80/100) \times (5/20) = 0.8$. The probability of the routine scenario (without any emergency) is 0.9, corresponding to normal operating conditions in the absence of typhoon shocks.

Each disruption scenario is also associated with a network resilience indicator, which reflects the system's ability to sustain or recover functionality under the given condition. The resilience level is determined by the intensity of the corresponding typhoon event. Specifically, disruption scenarios triggered by lower-intensity typhoons tend to have higher network resilience indicators due to milder impacts and faster recoverability, whereas those induced by more severe typhoons exhibit lower resilience. Based on existing studies (Byeon et al., 2020; Chen and Miller-Hooks, 2012; Wang et al., 2023), we assign the network resilience indicators as follows: 0.85 for TS, 0.8 for STS, 0.7 for TY, 0.6 for STY, and 0.5 for SuperTY. The routine scenario is assigned a baseline network resilience indicator of 0.9. On the demand side, we instantiate the uncertainty structure introduced in Section 4 by specifying the deviation of each OD pair as $\widehat{l}_n = \sigma \bar{l}_n$, where $\bar{l}_n \in [0.6, 0.8]$ and σ is a scaling coefficient. The total demand uncertainty budget Γ_ω is specified as $\Gamma_\omega = \xi_\omega |N|$, where $\xi_\omega \in [0, 1]$ is a fractional parameter referred to as the budget level, and $|N|$ denotes the total number of OD pairs. Γ_ω is a managerial policy parameter reflecting the platform's risk tolerance. A more conservative (risk-averse) platform may select a larger Γ_ω to protect against a larger aggregate level of demand estimation errors across OD pairs, whereas a less conservative platform may select a smaller Γ_ω to prioritize expected profit. In this study, Γ_ω is restricted to be an integer and varied to examine how different levels of demand uncertainty impact system performance.

6.2. Computational performance analysis

To evaluate the effectiveness of our proposed model, we conduct computational experiments with different values of the budget level parameter ξ, which controls the robustness of the network design. Specifically, we test four representative budget levels: $\xi = 0$, $\xi = 0.2$, $\xi = 0.5$, and $\xi = 0.8$, corresponding to budget values of 0, 2, 5, and 8, respectively. These values represent the average budget levels across all disruption scenarios. The experiments are based on the real-world case introduced in Section 6.1. The modeling framework, cost parameters, OD pairs, transportation arcs, disruption scenarios, and demand deviation factors remain consistent across all tests. The proposed model is solved using the customized CCG algorithm under a unified computational setting. To assess performance under increasing disruption severity, we construct three instance groups: ISG1 (6 disruption scenarios), ISG2 (12 disruption scenarios), and ISG3 (18 disruption scenarios). We set the deviation coefficient to $\sigma = 0.5$ to represent moderate demand volatility. The time limit for each run is set to 3600 s. When the time limit is reached, the current iteration is completed rather than forcibly terminated. The results are summarized in Table 2. As the uncertainty budget increases from 0 to 8, the objective value decreases consistently across all instances, reflecting the conservative nature of robust solutions under higher protection levels. On average, the profit drops by approximately 49% when the budget increases to 8, while the computation time rises increases from 26 s to 792 s.

Table 2
Comparisons of the results with different uncertainty budget.

Instance ID	$\Gamma = 0$		$\Gamma = 2$		$\Gamma = 5$		$\Gamma = 8$		GAP		
									Gap_1	Gap_2	Gap_3
	Z_0	T_0(s)	Z_1	T_1(s)	Z_2	T_2(s)	Z_3	T_3(s)	$\frac{Z_1 - Z_0}{Z_0}$	$\frac{Z_2 - Z_0}{Z_0}$	$\frac{Z_3 - Z_0}{Z_0}$
ISG1-1	1,471,387	5	1,027,095	55	820,640	125	728,939	185	30%	44%	50%
ISG1-2	2,578,275	3	1,993,783	32	1,664,628	161	1,522,761	177	23%	35%	41%
ISG1-3	780,430	2	510,475	87	362,327	122	351,302	132	35%	54%	55%
ISG2-1	1,817,322	15	1,313,845	369	1,149,722	649	1,138,555	780	28%	37%	37%
ISG2-2	1,965,548	24	1,466,187	387	1,121,751	572	1,052,525	712	25%	43%	46%
ISG2-3	1,169,506	18	671,553	343	637,611	581	574,688	566	43%	45%	51%
ISG3-1	938,084	54	624,886	507	407,632	972	346,212	1238	33%	57%	63%
ISG3-2	1,068,654	63	707,191	642	515,737	1011	492,235	1592	34%	52%	54%
ISG3-3	2,117,214	54	1,546,746	636	1,234,590	1206	1,123,964	1750	27%	42%	47%
Average		26		340		600		792	31%	45%	49%

Notes: (1) Instance ID is defined as "problem scale"-"index of instance", where "ISG1", "ISG2", and "ISG3" correspond with 6,12,18 disruption scenarios, respectively. (2) Z_0 and T_0 denote the objective value and computation time when uncertainty budget is zero, respectively. Z_0 indicates the nominal solution; Z_1 and T_1, Z_2 and T_2, Z_3 and T_3 correspond to objective values and computation times when uncertainty budget equals to two, five, and eight, respectively. (3) Gap_1, Gap_2, and Gap_3 represent the relative gaps between Z_0 and Z_1, Z_2, and Z_3, respectively.

6.3. Computational performance analysis of heuristic algorithm

This section investigates the computational performance of the CPU-GPU hybrid heuristic algorithm and compares it with the CCG algorithm. The experiments are conducted on four groups of instances: ISG4 (with 24 disruption scenarios), ISG5 (with 48 disruption scenarios), ISG6 (with 240 disruption scenarios), and ISG7 (with 480 disruption scenarios). The budget level is set to $\xi = 0.2$, corresponding to a budget value of 2; the deviation coefficient is set to $\sigma = 0.5$, representing moderate demand volatility. For the CCG algorithm, the termination criterion is either an elapsed time exceeding 7200 s or convergence of the upper and lower bounds, while for the heuristic algorithm, the termination criterion is that the number of iterations reaches 300. The results are shown in Table 3.

The experimental results indicate that the CCG algorithm converges within the time limit for ISG4, while for ISG5 it reaches the 7200-second limit and thus only provides the best feasible solution. For the larger-scale ISG6 and ISG7 instances, CCG fails to obtain feasible solutions within 7200 s; thus, the results of CCG for these instances are not reported. In contrast, the proposed hybrid heuristic algorithm consistently achieves feasible and high-quality solutions across all instances within short computation times. For ISG4 and ISG5, the average optimality gap between the heuristic algorithm and the CCG algorithm is 4.2%, demonstrating the reliability of the solution quality. In terms of computational efficiency, the CPU version heuristic algorithm reduces computation time by 99.5% on average compared to the CCG algorithm, highlighting a significant time advantage. Furthermore, by exploiting GPU parallelism, the CPU-GPU version further reduces runtime by 62.7% compared to the CPU version, achieving substantial acceleration on large-scale scenarios.

6.4. Managerial insights

By using the validated algorithm and model, sensitivity analyses are conducted to derive managerial insights that are potentially useful for 4PL platform operators in the real world.

6.4.1. Impact of city capacity disruption

To assess the structural vulnerabilities in the designed resilient multimodal network, we conduct a sensitivity experiment that evaluates how profit responds to capacity disruptions at individual cities. Specifically, for each tested city, we simulate local disruptions by reducing the remaining capacity of all its connected arcs to 85%, 80%, 75%, and 70%, while keeping other parameters unchanged. This allows us to quantify the nonlinear impact of localized disruptions on overall profit and identify cities whose disruptions have the most significant consequences.

Fig. 3 presents the profit reduction across four cities under varying remaining capacity levels. Results show that disruptions in Ningbo lead to the highest profit loss, while Xiamen causes the smallest impact. This indicates that different cities play asymmetric roles in maintaining network efficiency. Platform operators should classify cities like Ningbo as critical nodes and prioritize them in contingency planning. In practice, this could involve deploying redundant resources, setting more conservative capacity buffers, or establishing emergency rerouting plans to mitigate the amplification effect of their disruptions. Table 4 summarizes the number of connected rail, ship, and road arcs for each city, providing structural insight into these heterogeneous disruption impacts. Cities with fewer modal connections or lower redundancy exhibit greater vulnerability, which explains why disruptions at Ningbo produce

Table 3
Comparisons of the results of CCG and hybrid heuristic algorithm.

Instance ID	CCG		Hybrid heuristic algorithm			Optimality gap	Time comparison	
							Gap_{CPU}	Gap_{GPU}
	Z_C	T_C(s)	Z_H	T_H(s)	T_G(s)	$\frac{\lvert Z_C - Z_H\rvert}{Z_C}$	$\frac{\lvert T_C - T_H\rvert}{T_C}$	$\frac{\lvert T_H - T_G\rvert}{T_H}$
ISG4-1	748,306	4729	704,541	12	10	5.8%	99.7%	16.7%
ISG4-2	419,359	3839	413,840	19	12	1.3%	99.5%	36.8%
ISG4-3	651,324	3231	624,117	17	14	4.2%	99.5%	17.6%
ISG5-1	743,132	7200	732,278	45	16	1.5%	99.4%	64.4%
ISG5-2	898,336	7200	864,330	37	10	3.8%	99.5%	73.0%
ISG5-3	1,092,855	7200	998,015	36	14	8.7%	99.5%	61.1%
ISG6-1	N/A	N/A	1,045,473	95	16	N/A	N/A	83.2%
ISG6-2	N/A	N/A	631,028	69	19	N/A	N/A	72.5%
ISG6-3	N/A	N/A	469,340	99	18	N/A	N/A	81.8%
ISG7-1	N/A	N/A	1,068,598	156	25	N/A	N/A	84.0%
ISG7-2	N/A	N/A	253,121	139	24	N/A	N/A	82.7%
ISG7-3	N/A	N/A	1,025,123	129	28	N/A	N/A	78.3%
Average						4.2%	99.5%	62.7%

Notes: (1) Instance ID is defined as "problem scale"-"index of instance", where "ISG4", "ISG5", "ISG6", and "ISG7" correspond with 24, 48, 240, 480 disruption scenarios, respectively. (2) Z_C and T_C denote the objective value and computation time solved by the CCG algorithm, respectively. Z_H and T_H denote the objective value and computation time solved by the CPU version hybrid heuristic algorithm, respectively. T_G denote the computation time solved by the CPU-GPU version hybrid heuristic algorithm. (3) For ISG5 instances, the CCG procedure reaches the time limit of 7200 s before full convergence. Therefore, the reported Z_C is the best feasible solution found within 7200 s, and T_C is recorded as 7200 s.

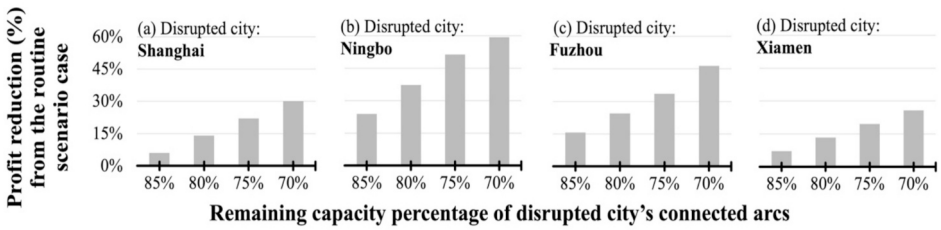


Fig. 3. Impact of remaining capacity percentage of arcs connected to disrupted cities on profit.

Table 4
The number of connected arcs for the four cities.

Disrupted cities	The number of connected rail arcs	The number of connected ship arcs	The number of connected road arcs
Shanghai	3	3	6
Ningbo	3	1	5
Fuzhou	3	2	5
Xiamen	1	2	6

substantially larger profit losses than disruptions at Xiamen. Because the structural factors in Table 4 appear to explain the observed impact patterns, we further verify this relationship using a regression model that predicts a "disruption-impact indicator" Z_{city} for each city: $Z_{city} = 7.25 \times 10^{-2} R_{rail} - 0.13 R_{ship} - 0.01 R_{road} + 1.83 rp$, where R_{rail}, R_{ship}, and R_{road} denote the number of connected rail, ship, and road arcs, respectively, and rp represents the remaining capacity percentage of connected arcs. The predicted indicator is close to the aforementioned profit reduction percentage for the disrupted cities in Fig. 3. The regression yields a high R-squared of 0.971 and a highly significant F-statistic ($F = 92.04$, $p = 2.22 \times 10^{-8}$), indicating excellent overall model fit and explanatory power. If the four cities in the real case undergo changes with respect to their connected arcs, platform operators could use the above formula to rapidly predict the potential impact of the disruption on each city's profit, identify the key cities, and take measures to reduce the negative impact of disruption. Note that the above formula is only applicable to the real case introduced in Section 6.1 and does not hold for all contexts; however, the methodology proposed in this paper can be used to derive the formulas for specific contexts.

6.4.2. Benefit of considering disruption scenarios

In multimodal network design, although extreme disruptions (e.g., typhoons) occur infrequently, their impact can be substantial on both profit and overall service performance. To evaluate the necessity of explicitly modeling disruptions, we conduct a comparative experiment. First, we assume the probability of the routine scenario is 1, thereby obtaining a solution without considering disruptions. Then, we evaluate this "no-disruption solution" under real disruption scenarios, while relaxing the resilience constraints to ensure feasibility. In the evaluation, the realized network resilience indicator, calculated as the ratio of fulfilled to nominal demand under disruptions, reflects actual performance and differs from the threshold values used as model constraints. The average network resilience indicator is computed as the weighted sum of realized network resilience indicators across all scenarios, using scenario probabilities as weights. We observe that the realized resilience indicators of the no-disruption solution drop sharply in more severe disruption scenarios, indicating a considerable decline in service performance. As shown in Table 5, the model that incorporates disruptions achieves significantly higher total profit and consistently better network resilience across scenarios. These results highlight that disruptions, even with low probability, can severely degrade the system if not properly modeled. Therefore, disruption-aware optimization is essential to maintain both economic performance and service reliability.

Table 5
Impact of considering disruption scenarios on network resilience indicator and profit.

Indicators	Model with disruption scenarios	Model without disruption scenarios
Realized NRI of Routine scenario	0.97	0.92
Realized NRI of TS scenario	0.93	0.71
Realized NRI of STS scenario	0.88	0.62
Realized NRI of TY scenario	0.8	0.55
Realized NRI of STY scenario	0.74	0.47
Realized NRI of SuperTY scenario	0.66	0.39
Average NRI	0.96	0.89
Total profit ($)	967,313	700,785
Profit increment due to considering disruption scenarios (percentage)	38%	

Note: "NRI" denotes the network resilience indicator.

6.4.3. Impact of parameter dispersion of OD pairs

In the real world, OD pairs differ in multiple characteristics. To assess how such difference in OD-specific characteristics influences solutions, we conduct a series of experiments by adjusting the dispersion (i.e., standard deviation) of three key parameters: maximum unit price (b_n), nominal demand scaled factor ($\bar{l}_n$), and maximum demand (u_n). In these experiments, based on the real case, we increase the standard deviation of each indicator by 10%, 20%, 30%, and 40% to simulate scenarios of increasing dispersion. To ensure experimental fairness, the mean values of these indicators are held constant, and only the standard deviations are varied.

In Fig. 4, the results show that increased standard deviation in both maximum unit price and the nominal demand scaled factor leads to improved profitability, with the latter having the most significant impact. This highlights the importance of precise estimation and continuous adjustment of demand scaled factor based on real-time data. In contrast, greater variability in maximum demand tends to reduce total profit. To mitigate it, 4PL platform operators are advised to avoid excessive disparities in maximum demand across OD pairs, such as by incorporating filtering mechanisms, prioritization policies, or strategic segmentation to avoid extreme disparities in demand levels of OD pairs.

7. Conclusion

This study proposes a decision model and a solution algorithm to support 4PL platforms in building resilient multimodal transportation networks under uncertainty. Based on the proposed scenario-based stochastic programming model embedded with robust optimization, the platform can jointly optimize three key decisions: service price for each OD pair, capacity planning for each arc, and capacity allocation decisions under different disruption scenarios. The first two decisions, namely service pricing and capacity planning, determine the structure of the multimodal transportation network. Once the network is established, the capacity allocation decision supports adaptive resource deployment under different disruption scenarios. The model aims to maximize the platform's profit, defined as the worst-case total revenue minus the capacity planning cost. Specifically, the total revenue is determined by the service prices of OD pairs and the actual demand fulfilled under different scenarios. The core contributions of this study can be summarized as follows:

From the perspective of problem modeling, this paper develops a scenario-based stochastic programming model embedded with robust optimization to systematically address the design of resilient multimodal transportation networks under both demand and supply uncertainties. The model integrates price-driven stochastic demand modeling, scenario-based arc capacity disruptions, and quantifiable resilience constraints, and adopts a max–min–max structure to evaluate worst-case profit guarantees. The framework is extensible and applicable to a broader class of network optimization and resource planning problems, particularly in decision environments characterized by (i) uncertainty on both the demand and supply sides; (ii) joint optimization of anticipatory planning and adaptive responses, with a typical two-stage or multi-stage decision structure; and (iii) the need to balance economic efficiency with robustness, emphasizing performance guarantees under uncertainty.

From the perspective of algorithm design, this paper develops a tailored CCG algorithm to solve the proposed model. To address the violation of the relatively complete recourse assumption, we incorporate a nested oracle structure. Specifically, a feasibility subproblem is first employed to identify scenarios in which pricing and capacity planning decisions cannot satisfy resilience constraints; once feasibility is confirmed across all scenarios, an optimality subproblem is then solved to determine the worst-case profit and generate valid cuts. Through computational tests under varying levels of uncertainty budget, we observe that increasing the uncertainty budget leads to a consistent decline in platform profit, indicating the conservative nature of robust solutions. Meanwhile, the computational time increases significantly, highlighting the trade-off between robustness and efficiency. To enhance scalability, especially for large-scale instances, we further develop a CPU-GPU hybrid heuristic. The method offloads the second-stage revenue evaluations to the GPU, enabling massively parallel processing of a large number of scenario realizations. Numerical results indicate that the hybrid approach achieves significant speedups compared with the CCG, while maintaining solution quality within a small optimality gap relative to exact algorithm.

From the perspective of managerial insights, we conduct sensitivity analyses to extract key operational guidelines for 4PL platforms: (i) *Identifying critical disruption nodes*: Disruptions in certain cities, such as Ningbo, lead to substantial profit loss, highlighting their importance in maintaining network resilience. Platform operators should prioritize these cities in contingency planning. (ii) *Benefit of modeling disruptions explicitly*: Modeling disruptions improves the platform's profit by 38% and enhances resilience across all scenarios. This confirms the necessity of disruption-aware optimization even under low-probability events. (iii) *Effect of OD-level parameter heterogeneity*: Moderate heterogeneity in price and demand factors enhances profitability, but excessive variation in maximum demand can reduce efficiency. Operators should avoid extreme disparities across OD pairs through segmentation or filtering.

Despite the effectiveness of the proposed model in supporting resilient multimodal network design under uncertainty, several modeling simplifications limit its generalizability. First, the demand function is assumed to be linear. Exploring alternative functional forms, such as exponential or logarithmic demand functions, may enhance the model's applicability. Second, this study focuses on network operations after disruptions but does not consider post-disruption recovery strategies. Future research could incorporate such recovery decisions into this framework to support both pre-disruption planning and post-disruption adaptation.

CRediT authorship contribution statement

Xiuwen Wang: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation.

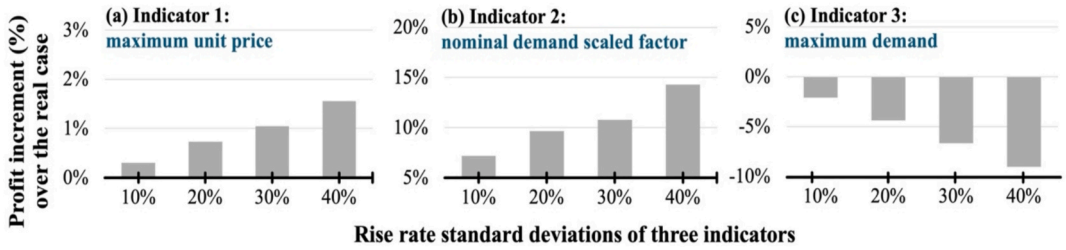


Fig. 4. Impact of parameter dispersion of OD pairs on profit.

Haiyang Shi: Writing – review & editing, Validation, Software, Methodology, Investigation. **Zifan Li:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Yu Wang:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation. **Lu Zhen:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Appendix. Notations

Sets and indices	
N	set of OD pairs, indexed by n.
R_n	set of transportation plans of OD pair n, indexed by r.
C	set of arcs, indexed by c.
Ω	set of scenarios, indexed by ω; it includes the routine scenario and several disruption scenarios.
$\mathscr{U}$	polyhedral uncertainty set for all scenarios.
$\mathscr{T}_\omega$	scenario-specific uncertainty set for scenario ω.
Γ_ω	the budget of uncertainty under scenario ω.
Parameters	
$f_n(x)$	transportation demand (TEUs) of OD pair n, if the unit price, which is quoted by the platform to customers, is x USD for transporting one TEU container for the OD pair.
v_{rc}	equal to one if transportation plan r uses arc c and zero otherwise.
w_c	unit renting cost (USD/TEU) of transportation capacity for arc c.
$g_{c\omega}$	percentage of remaining capacity on arc c under scenario ω; for the routine scenario, it equals one; for the disruption scenarios, its value is in interval (0,1).
p_ω	probability of scenario ω.
b_n	maximum unit price (USD/TEU) that could be quoted by the platform to customers for transporting one TEU container on OD pair n; if the unit price quoted by the platform is higher than b_n, the customers will resort to the 3PL carriers rather than the 4PL platform.
e_ω	threshold (the minimum value) for the network resilience indicator under scenario ω.
$\bar{l}_n$	nominal demand scaled factor.
$\widehat{l}_n$	maximum deviation of demand scaled factor.
$z_{n\omega}$	auxiliary deviation parameter representing the adversarial deviation of demand for OD pair n in scenario ω.
$\widetilde{l}_{n\omega}$	demand scaled factor of OD pair n under scenario ω, defined as $\widetilde{l}_{n\omega} = \bar{l}_n + z_{n\omega}\widehat{l}_n$.
Decision variables	
α_c	integer, number of rented TEUs of transportation capacity on arc c.
β_n	continuous, unit price (USD/TEU) quoted by the platform to customers for transporting one TEU container on OD pair n.
$\gamma_{r\omega}$	continuous, actual number of TEUs transported via transportation plan r under scenario ω.

Data availability

Data will be made available on request.

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